

Predictive Analysis of Caffeine Intake and Lifestyle Factors in University Students

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1. Introduction and Problem Description:

1.1. Problem Description

Lifestyle factors play an important role in influencing caffeine consumption and thereby providing health and behavioral insights among university students, who usually have an irregular sleep pattern, heavy workload, and various other external factors that might lead to increasing the consumption of caffeine intake. Our report examines the relation of caffeine intake with various factors like screen time, physical activity, sleep duration, study hours, and demographic data in terms of age, gender, and academic year.

The complex behavior of caffeine consumption is influenced by a wide range of environmental and lifestyle factors. However, the inherent variation in individual actions and the possibility of nonlinear correlations between the predictors and the outcome make the determination of these links extremely difficult to deduce reliably. Thus, a thorough analysis of various models is required to obtain one that can predict this complex behavior accurately.

1.2. Objective

The primary objective of our project is to develop predictive models that identify key factors influencing caffeine consumption among college students. By comparing the performance of various machine learning models, such as random forests, MLP and CNN neural networks, and linear regression, we aim to determine the most effective approach for predicting caffeine use based on the available data features and our interpretation of said features.

2. Data Description

2.1. File description

The dataset utilized for this study comprises records on sleep patterns, lifestyle factors, caffeine intake, and demographic information of university students. The data includes

14 variables collected from a sample population of 500 students, which we expect will help provide insights into the connection between various daily activities and caffeine intake. Below is an overview of all the variables included:

- **Student_ID:** Unique identifier for each student.
- **Age:** Age of the student.
- **Gender:** Gender of the student (e.g., Male, Female, Other).
- **University_Year:** Academic year of the student (e.g., 1st Year, 2nd Year).
- **Sleep_Duration:** Average hours of sleep per day.
- **Study_Hours:** Average hours spent studying daily.
- **Screen_Time:** Average hours of screen time per day.
- **Caffeine_Intake:** Number of caffeinated beverages consumed daily.
- **Physical_Activity:** Weekly minutes of physical activity.
- **Sleep_Quality:** Self-reported sleep quality score on a scale of 1 to 10.
- **Weekday_Sleep_Start:** Sleep start time on weekdays (in 24-hour format).
- **Weekend_Sleep_Start:** Sleep start time on weekends (in 24-hour format).
- **Weekday_Sleep_End:** Sleep end time on weekdays (in 24-hour format).
- **Weekend_Sleep_End:** Sleep end time on weekends (in 24-hour format).

2.2. Data Cleaning and Processing

The dataset was imported using Python's panda's library to facilitate preprocessing and subsequent analysis in the Python notebook Section 1.1 . The following steps were taken to ensure the data was clean and ready for analysis, as seen in python notebook Section 1.2:

- Calculation of Sleep Durations:** Weekend and weekday sleep durations were calculated based on the start and end times, accounting for scenarios where sleep might extend past midnight. The original start and end time columns were removed post-calculation.
- Removal of Irrelevant Features:** Columns such as `Student_ID` and `Sleep_Quality` were dropped. `Student_ID` was considered irrelevant for modeling, while `Sleep_Quality` was considered to be unsuitable as the definition of good sleep is subjective to each university student, making it classified as "junk data".
- Conversion of Categorical Variables:** Categorical columns such as `Gender` and `University_Year` were mapped to integer values to enable numerical processing.
- Simplification of Target Variable:** We recognize that ten bins aka ten integers representing the caffeine intake with only 500 data points does not result in accurate models from previous runs. Hence, we will be simplifying the caffeine intake column into integers 1, 2, 3 which will represent the intensity of caffeine intake, *Light*, *Medium*, and *Heavy*. Looking through the column of caffeine intake values, we

note that the values do not exceed 5 caffeinated drinks so we will indicate (1) light caffeine intake as 0 - 2 caffeinated drinks, (2) medium caffeine intake as 3 - 4 caffeinated drinks, and more than 4 caffeinated drinks as (3) heavy caffeine intake.

2.3. Inspection and Validation

After cleaning, we are left with a dataset that contains 500 samples with 10 features. The new list of features is: `Age`, `Gender`, `University_Year`, `Sleep_Duration`, `Study_Hours`, `Screen_Time`, `Caffeine_Intake`, `Physical_Activity`, `Weekend_Sleep_Duration`, `Weekday_Sleep_Duration`.

Box plots were generated in Section 1.3 of the notebook to visualize the relationships between caffeine intake intensity and screen time, as well as caffeine intake intensity and sleep duration.

In Section 1.4 of the notebook, the correlation matrix was computed to examine the relationships between the features and the target variable. The analysis revealed that weekday sleep duration had the highest correlation, with a value of 0.08. This suggests a weak relationship, indicating that weekday sleep duration likely provides minimal predictive power for the target variable. Also, this low correlation value indicates that caffeine intake is not strongly influenced by any single feature, with its factors having a relatively even distribution across multiple factors.

2.4. Variable Definition

Lastly, in Section 1.5, the processed dataset was divided into the feature set (X), which excludes the *Caffeine_Intake* column from the data, and the target variable (Y), which consists solely of the *Caffeine_Intake* column, for subsequent modeling.

3. Methodology

Linear Regression

This section details the methodology for implementing Linear Regression to predict caffeine intake based on various student lifestyle factors. The process is divided into two main approaches: a manual method utilizing Ordinary Least Squares (OLS) and an automated method employing LASSO regression with hyper parameter tuning.

The following *ordering* within the methodology section adheres to the sequence outlined in the Python notebook.

Python Notebook Section: 2.1 Linear Regression (Manual Method: Ordinary Least Squares)

Python Notebook Section: 2.1.1 Centering Inputs and Outputs

To simplify calculations and mitigate collinearity among features, the input features ($\mathbf{X}$)

and the target output ($\mathbf{Y}$) were centered:

$$X_c = X - \bar{X}, \quad Y_c = Y - \bar{Y}$$

where $\bar{X}$ and $\bar{Y}$ represent the mean of the input features and target output, respectively.

Python Notebook Section: 2.1.2 Calculating Parameters The centered data was used to compute the inverse of the input covariance matrix and estimate the regression parameters:

$$\theta_1 = (X_c^\top X_c)^{-1} X_c^\top Y_c$$

The intercept term, θ_0 , was calculated as:

$$\theta_0 = \bar{Y} - \bar{X} \cdot \theta_1$$

Python Notebook Section: 2.1.3 Making Predictions

The linear regression model made predictions ($\mathbf{Yhat}$) using:

$$Yhat = X \cdot \theta_1 + \theta_0$$

Python Notebook Section: 2.2 Linear Regression with Regularization (LASSO)

The Lasso class from the scikit-learn library module `linear_model` was utilized to apply a LASSO regularized linear regression.

Python Notebook Section: 2.2.1 Regularization Parameter Grid

The LASSO method incorporated regularization to reduce overfitting. A grid of regularization parameters (α) ranging from 10^{-2} to 10^2 was defined:

$$\alpha \in \{0.01, 0.1, 1, \dots, 100\}$$

Python Notebook Section: 2.2.2 Hyperparameter Tuning

GridSearchCV was employed to identify the best regularization parameter. The process used 5-fold cross-validation, optimizing the model for minimal negative mean squared error.

Python Notebook Section: 2.2.3 Best Alpha and Predictions

The optimal regularization parameter obtained was $\alpha = 2.07$. Using this value, predictions were made: ($\mathbf{Yhat_Lasso}$).

Neural Networks

This section outlines the methodology for building and evaluating neural networks to predict caffeine intake based on student lifestyle data. Two architectures are considered: a Multi-Layer Perceptron (MLP) and a Convolutional Neural Network (CNN). Each model is discussed in terms of data preprocessing, architecture, training, and evaluation.

Python Notebook Section: 3.1 Data Preprocessing

Neural networks require appropriately preprocessed data to ensure efficient learning. The preprocessing steps included:

- **Centering the Inputs:** Centering removes mean biases in features, ensuring faster convergence during training. Inputs were centered as follows:

$$X_c = X - \bar{X}$$

where $\bar{X}$ is the mean of the features. Our output does not need to be centered because the range does not correspond to a specific mean, as the neural network can learn to map the input appropriately to the desired output.

- **Scaling the Inputs:** Standard scaling was applied to normalize feature distributions, which allows better convergence.
- **One-Hot Encoding the Target Variable:** The categorical target variable (Caffeine Intake) was transformed into one-hot encoded labels to accommodate the softmax activation function seen later, ensuring the model learns probabilities for each class.

Python Notebook Section: 3.2 Multi-Layer Perceptron (MLP) Neural Network

Python Notebook Section: 3.2.1 Hyperparameter Tuning

GridSearchCV was used to determine the optimal hyperparameters for the MLP architecture. The hyperparameters included:

- **Batch Size:** [32, 64, 128]
- **Number of Epochs:** [50, 100]
- **Activation Function:** ['relu', 'tanh']
- **Dropout Rate:** [0.0, 0.2, 0.5]
- **Number of Neurons in Hidden Layer:** [64, 128, 256]
- **Optimizer:** ['adam', 'sgd']

A total of 648 combinations were evaluated using 3-fold cross-validation with accuracy as the scoring metric. The best parameters obtained were:

- Batch Size: 32
- Epochs: 50
- Activation: 'tanh'
- Dropout Rate: 0.2
- Neurons: 256
- Optimizer: 'adam'

Python Notebook Section: 3.2.2 Model Architecture and Training

The selected MLP architecture consisted of a dense layer with 256 neurons and a 'tanh' activation function, along with dropout regularization at a rate of 0.2 to prevent overfitting. The model was trained for 50 epochs using a batch size of 32, with 10 percent of the data reserved for validation.

Python Notebook Section: 3.3 Convolutional Neural Network (CNN)

Python Notebook Section: 3.3.1 Data Reshaping

Since CNNs require multidimensional inputs, which we saw in the letter image processing demo, the scaled feature matrix was reshaped to 3D to accommodate for a 1D convolution.

Python Notebook Section: 3.3.2 Model Architecture and Training

We want to see if the CNN can perform better than the MLP using the same parameters, so we will not be running the GridSearchCV again. If we wanted to see the most optimal parameters specific for a CNN, we could run GridSearch again specifically for a CNN model. Thus, the CNN architecture had the same "best parameters" used previously for the MLP.

Random Forests

Random forests was chosen for its ability to handle complex relationships and interactions between features.

Python Notebook Section: 4.1 Data Splitting

The dataset was split into training and testing sets to evaluate model performance. Specifically:

- **Training Set:** 90% of the data was used to train the model.
- **Testing Set:** The remaining 10% was reserved for model evaluation.

This ensures the model generalizes well to unseen data. The split was performed using the `train_test_split` function from scikit-learn with a random seed of 2382 for reproducibility.

Python Notebook Section: 4.2 Hyperparameter Tuning

GridSearchCV was used to sweep the number of trees for the Random Forest model.

- **Number of Trees (`n_estimators`):** Ranged from 10 to 500 in 50 equally spaced intervals.
- **Maximum Features (`max_features`):** Options were 'sqrt' (square root of the number of features) and 'log2' (logarithm base 2 of the number of features).

A total of 100 combinations were evaluated using 3-fold cross-validation with accuracy as the scoring metric. The best parameters identified were:

- `n_estimators`: 10
- `max_features`: 'sqrt'

Python Notebook Section: 4.3 Model Training

The Random Forest model was trained on the set using the best parameters identified during hyperparameter tuning. Random Forests combine multiple decision trees, each trained on a random subset of the data, and aggregate their predictions to improve accuracy and reduce overfitting.

Performance Evaluation

For all the previously state models, the model's performance was evaluated using the following metrics:

- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{\sum(Y - Y_{hat})^2}{\sum(Y - \bar{Y})^2}$$

- **Mean Squared Error (MSE):**

$$MSE = \frac{1}{N} \sum (Y - Y_{hat})^2$$

- **Mean Absolute Error (MAE):**

$$MAE = \frac{1}{N} \sum |Y - Y_{hat}|$$

4. Results and Analysis

This section presents a comprehensive analysis of the results obtained from all models, including Linear Regression, Neural Networks, and Random Forests. The performance of each model is evaluated using R^2 , MSE , and MAE .

4.1 Linear Regression Results

Linear regression was evaluated using two approaches: Ordinary Least Squares (OLS) and LASSO with hyperparameter tuning.

4.1.1 Ordinary Least Squares (OLS)

- **Coefficient of Determination (R^2):** 0.023. This indicates that only 2.3% of the variance in caffeine intake is explained by the model, suggesting a weak relationship between the features and caffeine intake.

- **Mean Squared Error (MSE):** 0.515. This value represents moderate prediction errors.
- **Mean Absolute Error (MAE):** 0.639. This highlights noticeable prediction inaccuracies.

4.1.2 LASSO Regression

- **Coefficient of Determination (R^2):** -5.073. This negative value indicates that the model performs worse than a simple mean prediction.
- **Mean Squared Error (MSE):** 3.204. This reflects substantial prediction errors.
- **Mean Absolute Error (MAE):** 1.636. This suggests significant inaccuracies in the model's predictions.

Conclusion: Both OLS and LASSO performed poorly, indicating that linear relationships do not adequately capture the complexities of the data.

4.2 Neural Network Results

Two types of neural networks were explored: Multi-Layer Perceptron (MLP) and Convolutional Neural Network (CNN).

4.2.1 Multi-Layer Perceptron (MLP)

- **Coefficient of Determination (R^2):** -0.760. This negative value indicates poor model performance.
- **Mean Squared Error (MSE):** 0.515. This reflects moderate errors similar to OLS.
- **Mean Absolute Error (MAE):** 0.639. Prediction inaccuracies remain significant.

4.2.2 Convolutional Neural Network (CNN)

- **Coefficient of Determination (R^2):** -0.437. This negative value indicates a poor fit to the data.
- **Mean Squared Error (MSE):** 0.758. This suggests higher errors compared to the MLP.
- **Mean Absolute Error (MAE):** 0.538. While slightly better than MLP, inaccuracies persist.

Conclusion: Both neural network models underperformed, failing to capture significant relationships in the data. The negative R^2 values highlight their inability to generalize.

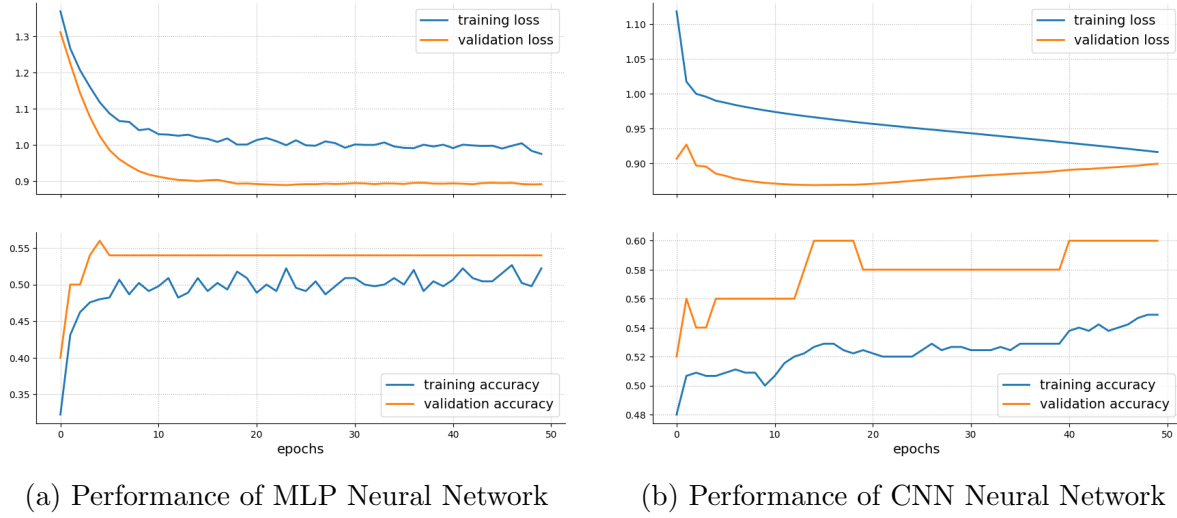


Figure 1: Comparison of Neural Network Performance

From Figure 1 (a), we can see that the MLP neural network demonstrates a validation loss stabilizing early and maintaining a small gap from training loss, while validation accuracy remains higher than training accuracy throughout training, which suggests no significant overfitting. From this we can see for future runs, stopping around 20 epochs is ideal as metrics stabilize with diminishing returns in later epochs.

From Figure 1 (b), we can see that the CNN neural network shows signs of overfitting, as the validation loss decreases initially but starts increasing after around 10 epochs, while the training loss continues to decrease steadily. Validation accuracy plateaus and fluctuates slightly, remaining higher than training accuracy. From this we can see for future runs, stopping around 10 epochs which could help prevent further overfitting and stabilize performance.

4.3 Random Forest Results

Random Forests outperformed all other models, demonstrating strong predictive capability.

- **Coefficient of Determination (R^2):** 0.757. This indicates that 75.7% of the variance in caffeine intake is explained by the model.
- **Mean Squared Error (MSE):** 0.128. This low value highlights minimal prediction errors.
- **Mean Absolute Error (MAE):** 0.096. This reflects high accuracy in predictions.

Conclusion: Random Forests effectively captured the relationships between features and caffeine intake, outperforming linear and neural network models.

4.4 Overall Analysis and Data Insights

This analysis sheds light on the dataset and how the models performed. The dataset looks at factors influencing caffeine intake among university students, including variables like sleep duration, study hours, and screen time. These individual variables had weak correlations with caffeine intake—weekday sleep duration, for example, only had a correlation of 0.08—suggesting that no single factor is a strong predictor on its own.

The Random Forest model stood out, achieving an impressive R^2 of 0.757 by capturing complex interactions between variables. This highlights the importance of looking at how factors work together rather than individually. For instance, while screen time and study hours are weak predictors on their own, their combined effects might paint a clearer picture of caffeine consumption.

On the other hand, linear models struggled because they couldn't capture these interactions, and neural networks didn't perform well either, likely because of the small dataset and suboptimal tuning. This shows that neural networks need both a large amount of data and careful optimization.

In summary, caffeine intake is influenced by how multiple factors interact, not just by individual variables. Future efforts could focus on understanding these interactions better, using larger datasets to improve model performance.